

Packet Tracer - Modify Single-Area OSPFv2

Addressing Table

Device

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.16.1.1	255.255.255.0	N/A
	S0/0/0	172.16.3.1	255.255.255.252	
	S0/0/1	192.168.10.5	255.255.255.252	
R2	G0/0	172.16.2.1	255.255.255.0	N/A
	S0/0/0	172.16.3.2	255.255.255.252	
	S0/0/1	192.168.10.9	255.255.255.252	
	S0/1/0	209.165.200.225	255.255.255.224	
R3	G0/0	192.168.1.1	255.255.255.0	N/A
	S0/0/0	192.168.10.6	255.255.255.252	
	S0/0/1	192.168.10.10	255.255.255.252	
PC1	NIC	172.16.1.2	255.255.255.0	172.16.1.1
PC2	NIC	172.16.2.2	255.255.255.0	172.16.2.1
PC3	NIC	192.168.1.2	255.255.255.0	192.168.1.1
Web Server	NIC	64.100.1.2	255.255.255.0	64.100.1.1

Objectives

Part 1: Modify OSPF Default Settings

Part 2: Verify Connectivity

Scenario

In this activity, OSPF is already configured and all end devices currently have full connectivity. You will modify the default OSPF routing configurations by changing the hello and dead timers and adjusting the bandwidth of a link. Then you will verify that full connectivity is restored for all end devices.

Instructions

Part 1: Modify OSPF Default Settings

Step 1: Test connectivity between all end devices.

Before modifying the OSPF settings, verify that all PCs can ping the web server and each other.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Request timed out.
Reply from 172.16.2.2: bytes=32 time=6ms TTL=126
Reply from 172.16.2.2: bytes=32 time=5ms TTL=126
Reply from 172.16.2.2: bytes=32 time=4ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 6ms, Average = 5ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.2: bytes=32 time=4ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126
Reply from 192.168.1.2: bytes=32 time=5ms TTL=126

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 5ms, Average = 4ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Request timed out.
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Reply from 64.100.1.2: bytes=32 time=7ms TTL=125
Reply from 64.100.1.2: bytes=32 time=7ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 7ms, Average = 6ms

C:\>
```

OSPF ทำงานปกติ network ยังคงใช้ได้

Step 2: Adjust the hello and dead timers between R1 and R2.

a. Enter the following commands on R1.

Open configuration window

```
R1(config)# interface s0/0/0
```

```
R1(config-if)# ip ospf hello-interval 15
```

```
R1(config-if)# ip ospf dead-interval 60
```

b. After a short period of time, the OSPF connection with R2 will fail, as shown in the router output.

00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

```
R1#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.16.1.1      YES manual up          up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/0/0              172.16.3.1      YES manual up          up
Serial0/0/1              192.168.10.5    YES manual up          up
Serial0/1/0              unassigned      YES unset  administratively down down
Serial0/1/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down

R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#ip ospf hello-interval 15
R1(config-if)#ip ospf dead-interval 60
R1(config-if)#
12:14:50: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Dead timer expired

12:14:50: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
```

00:02:40: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached

```
12:14:50: %OSPF-5-ADJCHG: Process 1, Nbr 209.165.200.225 on Serial0/0/0 from FULL to DOWN, Neighbor Down: Interface down or detached
```

Both sides of the connection need to have the same timer values in order for the adjacency to be maintained. Identify the interface on R2 that is connected to R1. Adjust the timers on the R2 interface to match the settings on R1.

After a brief period of time you should see a status message that indicates that the OSPF adjacency has been reestablished.

00:21:52: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done

```
R2>enable
R2#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       172.16.2.1      YES manual up          up
GigabitEthernet0/1       unassigned      YES unset  administratively down down
GigabitEthernet0/2       unassigned      YES unset  administratively down down
Serial0/0/0              172.16.3.2      YES manual up          up
Serial0/0/1              192.168.10.9    YES manual up          up
Serial0/1/0              209.165.200.225 YES manual up          up
Serial0/1/1              unassigned      YES unset  administratively down down
Vlan1                    unassigned      YES unset  administratively down down
R2#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#interface s0/0/0
R2(config-if)#ip ospf hello-interval 15
R2(config-if)#ip ospf dead-interval 60
R2(config-if)#
12:23:33: %OSPF-5-ADJCHG: Process 1, Nbr 192.168.10.5 on Serial0/0/0 from LOADING to FULL, Loading Done
```

Step 3: Adjust the bandwidth setting on R1.

a. Trace the path between PC1 and the web server located at 64.100.1.2. Notice that the path from PC1 to 64.100.1.2 is routed through R2. OSPF prefers the lower cost path.

C:\> tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

1 1 ms 0 ms 8 ms 172.16.1.1

2 0 ms 1 ms 0 ms 172.16.3.2

3 1 ms 9 ms 2 ms 209.165.200.226

4 * 1 ms 0 ms 64.100.1.2

Trace complete.

```

C:\>tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    172.16.1.1
  2  2 ms    0 ms    2 ms    172.16.3.2
  3  1 ms    5 ms    7 ms    209.165.200.226
  4  0 ms    0 ms    1 ms    64.100.1.2

Trace complete.

C:\>

```

b. On the R1 Serial 0/0/0 interface, set the bandwidth to 64 Kb/s. This does not change the actual port speed, only the metric that the OSPF process on R1 will use to calculate best routes.

R1(config-if)# bandwidth 64

```

R1>enable
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface s0/0/0
R1(config-if)#bandwidth 64
R1(config-if)#

```

คำสั่งนี้ ไม่ใช่การปรับความเร็วจริง ของสาย (มันยังวิ่งที่ความเร็ว 1.544 Mbps หรือ T1 อยู่) แต่มันเป็นการบอก OSPF ว่า "ให้คิดว่าเส้นนี้มันช้าแค่ 64 Kbps" เพื่อให้ Cost มันสูงขึ้น ใช้ในการคำนวณค่า Cost

c. Trace the path between PC1 and the web server located at 64.100.1.2. Notice that the path from PC1 to 64.100.1.2 is redirected through R3. OSPF prefers the lower cost path.

```
C:\> tracert 64.100.1.2
```

Tracing route to 64.100.1.2 over a maximum of 30 hops:

```

1 1 ms 0 ms 3 ms 172.16.1.1
2 8 ms 1 ms 1 ms 192.168.10.6
3 2 ms 0 ms 2 ms 172.16.3.2
4 2 ms 3 ms 1 ms 209.165.200.226

```

5 2 ms 11 ms 11 ms 64.100.1.2

Trace complete.

```
C:\>tracert 64.100.1.2

Tracing route to 64.100.1.2 over a maximum of 30 hops:

  1  0 ms    0 ms    0 ms    172.16.1.1
  2  1 ms    2 ms    2 ms    192.168.10.6
  3  2 ms    7 ms    1 ms    172.16.3.2
  4  1 ms    4 ms    5 ms    209.165.200.226
  5  1 ms    0 ms    6 ms    64.100.1.2

Trace complete.
```

Close configuration window

PC1 -> R1 -> R3 -> R2 -> Web Server

ทำไมถึงเปลี่ยน? เพราะ R1 เห็นว่าเส้นทางผ่าน Serial 0/0/0 (ไปหา R2 โดยตรง) มี Cost สูงกว่าผ่าน R3 จึงเลือกไปทาง R3 แทน (ถึงจะอ้อมแต่ OSPF คิดว่ามันดีกว่า)

Part 2: Verify Connectivity

Verify that all PCs can ping the web server and each other.

PC3

Desktop Programming

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.16.1.2

Pinging 172.16.1.2 with 32 bytes of data:

Reply from 172.16.1.2: bytes=32 time=5ms TTL=126
Reply from 172.16.1.2: bytes=32 time=5ms TTL=126
Reply from 172.16.1.2: bytes=32 time=6ms TTL=126
Reply from 172.16.1.2: bytes=32 time=6ms TTL=126

Ping statistics for 172.16.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 6ms, Average = 5ms

C:\>ping 172.16.2.2

Pinging 172.16.2.2 with 32 bytes of data:

Reply from 172.16.2.2: bytes=32 time=5ms TTL=126
Reply from 172.16.2.2: bytes=32 time=2ms TTL=126
Reply from 172.16.2.2: bytes=32 time=6ms TTL=126
Reply from 172.16.2.2: bytes=32 time=6ms TTL=126

Ping statistics for 172.16.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 6ms, Average = 4ms

C:\>ping 64.100.1.2

Pinging 64.100.1.2 with 32 bytes of data:

Reply from 64.100.1.2: bytes=32 time=10ms TTL=125
Reply from 64.100.1.2: bytes=32 time=9ms TTL=125
Reply from 64.100.1.2: bytes=32 time=6ms TTL=125
Reply from 64.100.1.2: bytes=32 time=8ms TTL=125

Ping statistics for 64.100.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 6ms, Maximum = 10ms, Average = 8ms

C:\>|
```

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ข้อสงสัยส่วนตัวจากแลปนี้

ทำไมต้องปรับ Timer?

น่าจะเป็นเพราะ ในโลกความจริง บางที link มันช้าหรือมี Latency สูง ถ้าใช้ค่า Default (10/40) OSPF อาจคิดว่าเพื่อนบ้านตายบ่อยๆ (Flapping) การปรับค่าให้สูงขึ้น (เช่น 15/60) ช่วยให้ Network เสถียรขึ้นบน Link ที่ช้าลงได้นั่นเอง

ทำไมไม่ปรับ Bandwidth จริง?

เพราะเราไม่สามารถไปเปลี่ยนความเร็วสาย T1 จริงๆ ได้ (มันคือสัญญา ISP) แต่การโกงค่า Bandwidth ใน Router ช่วยให้เราสามารถ "บังคับ" OSPF ให้เลือกเส้นทางสำรองได้ โดยไม่ต้องไปยุ่งกับ Hardware

Cost คำนวณยังไง?

โดยปกติ OSPF Cost = Reference Bandwidth (100 Mbps) / Interface Bandwidth

เช่น ถ้า BW = 64 Kbps Cost จะเท่ากับ $100,000 / 64 = 1562$ (ซึ่งสูงมาก) ทำให้ OSPF เลี่ยงเส้นทางนี้ คือเราไปหลอกมันให้มันเปลี่ยนเส้นทาง